

Mubashir Rehman

AI Backend Engineer · Python / FastAPI / Django · LLM Integration & Agents · pgvector · PostgreSQL / Docker
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SUMMARY

Lead Software Engineer with **3+ years** building production Python backends and the AI layers on top: LLM agent tool-calling, pgvector semantic search, self-hosted inference. Co-author of a peer-reviewed Springer paper on ML (CSSP, 2025).

PUBLICATION

Evaluation of a low-cost single-lead ECG module for vascular ageing prediction and studying smoking-induced changes in ECG

Circuits, Systems, and Signal Processing (Springer), 44(8), 5852-5875, 2025 · DOI: 10.1007/s00034-025-03048-2

Co-author, equal contribution (3rd of 7 authors): data collection and the ML pipeline. **42 subjects, 6,131 segmented samples**; random forest **R² 0.99** on segmented data and **0.87** with transfer learning on unsegmented data.

EXPERIENCE

Lead Software Engineer, TransData · Lahore, Pakistan · Mar 2025 to Present (leading 5 engineers since Mar 2026)

- Built a crypto and social-intelligence backend end to end as sole author (**154 of 155 commits**) and **handed it over to the client**, by writing Telegram, Discord and Twitter/X ingestion, AI summarisation, pgvector semantic search on Gemini embeddings, and an alert engine with ARQ background tasks on Redis.
- Got self-hosted LLM inference **adopted by the engineering team**, by deploying vLLM on a single 12 GB GPU behind a LiteLLM OpenAI-compatible proxy and benchmarking open-weight models to select an AWQ-quantised 8B model.
- Built an AI-agent ERP platform (**two live deployments, demo and production**), by writing LLM agents with forced tool-calling against ERPNext through one versioned contract, a multi-company resolver (company, currency, warehouse per order), and purchase-order drafting and approval with audit logging.
- Implemented **10+ of 28 spec'd AI features** (27 of 27 commits, sole author) for a property-management platform, by building tenant scoring, churn and renewal prediction, sentiment analysis, document classification and LLM reply drafting behind an OpenAI and AWS Bedrock switching layer, live-tested against the OpenAI API.
- Built a Claude Code workflow for a voice-AI product (**11 project skills, 3 subagents**, each written after an incident), by splitting read-only production analysis from human-approved fixes, with a stronger reviewer model gating decisions.
- Built an AI voice receptionist for dental clinics through its **production launch**, on AWS infrastructure built from scratch that became its production backend, by wiring voice agents to booking and eligibility tools across three providers.
- Let prompt changes ship without a code deploy (**28 prompts across 11 workflows** migrated, 191 of 268 commits), by moving an LLM content platform's prompts out of an external workflow tool into an in-house, versioned prompt system.

Software Engineering Team Lead, VeritusLabs · Oct 2023 to Mar 2025 (Software Engineer Jul 2023 to Oct 2023)

- Promoted to Team Lead and led **a team of 4 engineers** on QGroundControl-based drone-control and swarm features in C++ and Qt/QML, including mission planning and real-time telemetry over MQTT.
- Delivered a full-stack AI prompt-management web app with streaming OpenAI responses, RBAC and S3-backed storage, **deployed through Docker and CI/CD**, by building its Django/FastAPI and LangChain backend.

PROJECTS

HireTrack · React 19, TypeScript, LangGraph, Supabase · github.com/mubashir-rehman/job-application-tracker

- Built and open-sourced a job-application tracker (sole author, **89 commits, public repo, live deployment**), by shipping a deterministic-first LangGraph pipeline that tailors resumes and scores job descriptions across five AI providers (bring-your-own-key), plus an offline-first PWA and a remote MCP server.

SKILLS

AI/LLM: OpenAI API, Gemini, LangChain, LangGraph, agent tool-calling, MCP, embeddings, pgvector, semantic search, vLLM, LiteLLM, Claude Code, MindsDB, Databricks, ElevenLabs, Retell, Telnyx

Languages: Python, SQL, TypeScript, JavaScript, C++, Bash

Backend and data: Django, Django REST Framework, FastAPI, Node.js, REST API design, OAuth2, JWT, RBAC, PostgreSQL, MySQL, Redis, SQLAlchemy, Alembic, Celery, ARQ

DevOps: Docker, Nginx, Traefik, GitHub Actions, CI/CD, AWS (ECS/Fargate, RDS, S3, IAM, KMS), Linux

EDUCATION

BS Computer Science, Information Technology University, Lahore